EXCEL ACADEMY

CLASS – 10 (CBSE)/ SCIENCE – HEREDITY NOTES

Introduction to Heredity

- Reproduction leads to new individuals that are similar but not identical to their parents.
- Variations occur during reproduction, more so in sexual reproduction compared to asexual reproduction.
- For example, sugarcane plants (asexual reproduction) show little variation, while sexually reproducing animals like humans show distinct variations.
- These variations are crucial for survival and evolution as the environment selects individuals with advantageous traits.

Accumulation of Variation During Reproduction

- New generations inherit traits with some variations from parents and also develop new differences.
- In asexual reproduction, offspring are nearly identical with minor variations due to DNA copying errors.
- Sexual reproduction maximizes variation among offspring.
- Not all variations survive; environmental factors select the variants best suited for survival.

Mendel's Contributions to Heredity

- Gregor Mendel, through his experiments on pea plants, discovered laws of inheritance over a century ago.
- Mendel studied traits like tall/short plants, round/wrinkled seeds, and flower colors.
- He observed that traits are inherited in predictable patterns rather than blended.

Mendel's Experiment on Tall and Short Plants

- Crossing tall (TT) and short (tt) plants resulted in all tall plants in the first generation (F1).
- In the second generation (F2), about 3/4 were tall and 1/4 were short.
- This showed two copies of a gene control traits—one from each parent.
- Dominant traits (e.g., tallness) express when present, while recessive traits (e.g., shortness) express only if both gene copies are recessive.

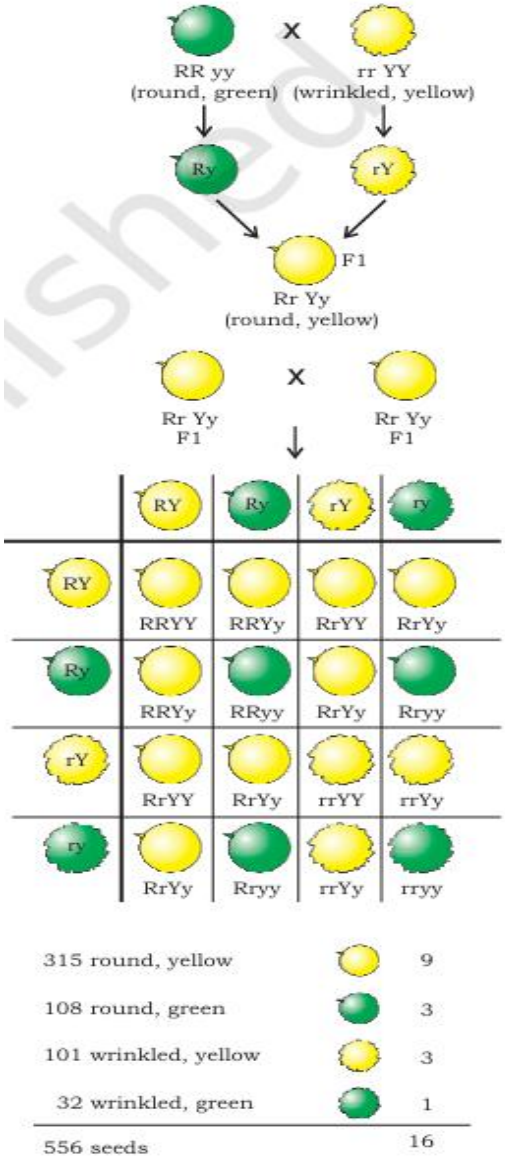
Diagram: Inheritance of Tall and Short Traits (Fig. 8.3)

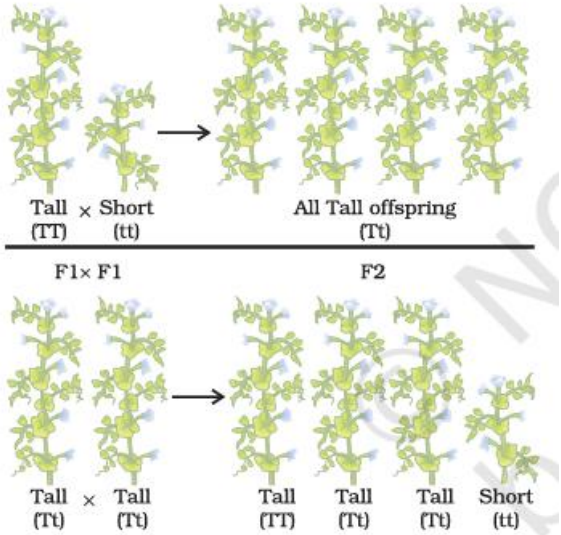
- Displays how two gene copies (TT, Tt, tt) affect plant height in offspring across generations.

Independent Inheritance of Traits

- Mendel also showed traits like seed shape (round or wrinkled) and color (yellow or green) are inherited independently.
- Crossing tall, round-seeded plants with short, wrinkled-seeded plants led to progeny with new trait combinations.
- This demonstrated independent segregation and recombination of gene pairs during gamete formation.

Diagram: Independent Inheritance of Seed Traits (Fig. 8.5)





- Shows distribution of seed shape and color traits in the offspring.

How Traits Get Expressed

- Genes are sections of DNA coding proteins, which control characteristics.
- For instance, plant height depends on concentrations of a growth hormone triggered by an enzyme.
- The efficiency of this enzyme, controlled by genes, determines the trait expressed.
- Both parents contribute equally to the DNA of offspring; thus, offspring inherit two gene copies per trait.

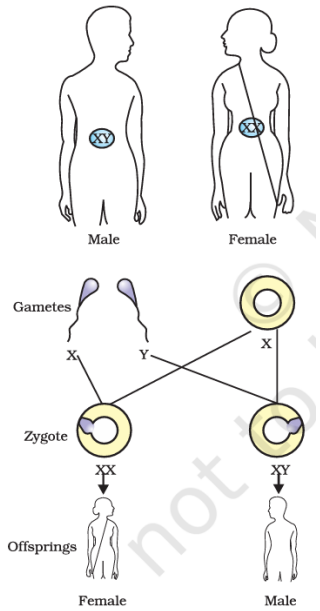
Chromosomes and Gene Sets

- Each organism has chromosomes (independent DNA pieces).
- Somatic cells have pairs of chromosomes (one from each parent).
- Germ cells (gametes) carry only one set of chromosomes.
- Sexual reproduction restores the paired chromosome number.

Sex Determination in Humans

- Human sex is genetically determined by sex chromosomes: females have XX, males XY.
- Fathers determine the sex of offspring by contributing either an X or Y chromosome.
- Half the children are expected to be male (XY) and half female (XX).

Diagram: Human Sex Determination (Fig. 8.6)



- Illustrates the chromosome process leading to male or female offspring.

Important Terms

- **Dominant Trait:** A trait that shows up even if only one copy of the gene is present (e.g., tallness).
- **Recessive Trait:** A trait that shows up only when both gene copies are recessive (e.g., shortness).
- **Gene:** A DNA segment coding for a protein influencing a trait.
- **Chromosome:** Thread-like DNA structure carrying many genes.
- **Gamete:** Sex cell carrying one set of chromosomes.

Summary of Key Points for CBSE Exams

- Variations arise in reproduction, driving evolution and survival.
- Mendel's laws explain how traits are inherited through dominant and recessive alleles.
- Traits can be inherited independently leading to new combinations.
- DNA and chromosomes are fundamental to heredity.
- Sex in humans is determined by X and Y chromosomes contributed by parents.